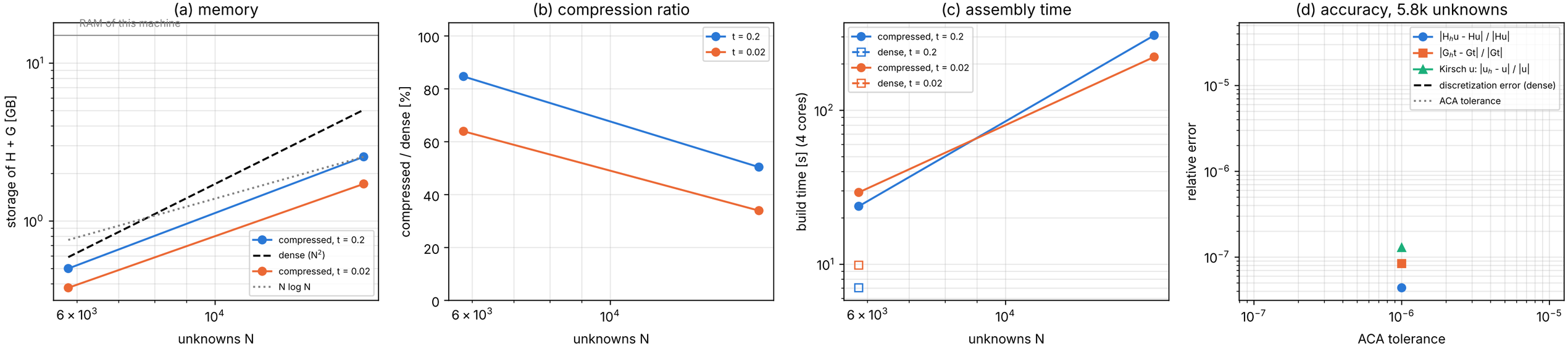
(a) memory
RAM of this machine
storage of H + G [GB]
$10^1$
$10^0$
compressed, t = 0.2
dense ($N^2$)
compressed, t = 0.02
N log N
$6 \times 10^3$
$10^4$
unknowns N
(b) compression ratio
compressed / dense [%]
100
80
60
40
20
0
t = 0.2
t = 0.02
$6 \times 10^3$
$10^4$
unknowns N
(c) assembly time
build time [s] (4 cores)
$10^2$
$10^1$
compressed, t = 0.2
dense, t = 0.2
compressed, t = 0.02
dense, t = 0.02
$6 \times 10^3$
$10^4$
unknowns N
(d) accuracy, 5.8k unknowns
relative error
$10^{-5}$
$10^{-6}$
$10^{-7}$
$|H_h u - Hu| / |Hu|$
$|G_h t - Gt| / |Gt|$
Kirsch u: $|u_h - u| / |u|$
discretization error (dense)
ACA tolerance
$10^{-7}$
$10^{-6}$
$10^{-5}$
ACA tolerance